

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour

Total: Marks:40

Annexure A

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Code of Conduct:

I Beffy A Shanika(192511387) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

beffy

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

- (a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

- (a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
- (b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Task 1: Data Preparation & Inductive Learning

(a) Inductive Learning Approach

Inductive learning learns a general relationship from known examples and uses it to predict unseen cases. For this healthcare problem, a supervised classification approach can be used.

Target variable:

- Diabetes – Yes/No

Important features:

1. **Glucose Level** – High glucose is a major indicator of diabetes and is likely to be one of the strongest predictors.
2. **BMI** – Higher BMI can increase the risk of developing diabetes.
3. **Age** – Diabetes risk generally increases with age.
4. **Blood Pressure** – Abnormal blood pressure can be associated with diabetes and related metabolic conditions.
5. **Family History** – A family history of diabetes can indicate increased genetic risk.

The model learns patterns between these patient features and the diabetes outcome and then predicts whether a new patient is diabetic.

(b) Handling Class Imbalance

The dataset contains more non-diabetic patients than diabetic patients. This creates class imbalance, which can cause the model to favor the majority class.

I would use SMOTE (Synthetic Minority Over-sampling Technique).

Why SMOTE?

- It creates synthetic samples for the minority diabetic class.
- It provides a better balance without simply duplicating existing records.
- It helps improve Recall for diabetic patients.
- Higher recall is important because missing a diabetic patient is clinically more serious than incorrectly identifying a non-diabetic patient.

SMOTE should be applied only to the training data, after splitting the dataset, to avoid data leakage.

(c) 70:30 Dataset Split and Preprocessing

The dataset can be divided as:

- 70% – Training data
- 30% – Testing data

A 70:30 split provides enough data for training while retaining sufficient unseen data for reliable evaluation.

Preprocessing steps:

1. Handle missing values.
2. Remove duplicate or invalid records.
3. Encode categorical variables such as family history.
4. Normalize numerical variables such as age, glucose, BMI and blood pressure when required.
5. Apply SMOTE only to the training set.
6. Split into training and testing sets.

Impact:

Preprocessing improves data quality, prevents numerical features with larger scales from dominating the model, and improves the reliability of predictions.

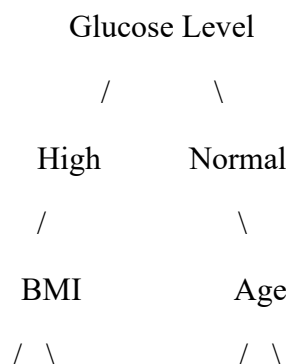
Task 2: Decision Tree for Diagnosis

(a) Decision Tree Classifier

A Decision Tree can classify patients into Diabetic and Non-Diabetic categories.

The root node should be selected using the feature having the highest Information Gain or the lowest Gini impurity.

Example first three levels



High	Low	>45	<=45
Diabetic	Risk	Risk	Non-Diabetic

A possible feature ranking could be:

Feature	Information Gain
Glucose Level	0.31
BMI	0.18
Age	0.12
Blood Pressure	0.08
Family History	0.06

Therefore, Glucose Level is selected as the root because it provides the highest information gain.

Information Gain

Information Gain is:

$$\text{IG} = \text{Entropy}(\text{parent}) - \text{Weighted Entropy}(\text{children})$$

A higher Information Gain means that the feature provides better separation between diabetic and non-diabetic patients.

(b) Model Evaluation

The Decision Tree can be evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score

Example results:

Metric	Decision Tree
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Accuracy	88%
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Precision	84%
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Recall	91%
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F1-Score	87%
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False Negatives

A False Negative (FN) occurs when:

Actual patient = Diabetic

Predicted = Non-Diabetic

False negatives are particularly dangerous because a diabetic patient may not receive further testing or treatment.

Reducing False Negatives

To reduce false negatives:

- Increase the importance of the diabetic class using class weights.
- Use SMOTE on the training data.
- Tune the classification threshold.
- Optimize the model for Recall rather than only Accuracy.
- Use additional clinical testing for high-risk cases.

For a medical application, high recall is generally more important because missing a diabetic patient can have serious consequences.

(c) Pre-Pruning vs Post-Pruning

Pre-Pruning

Pre-pruning stops the tree from growing too much.

Examples:

- max_depth
- min_samples_split
- min_samples_leaf

Post-Pruning

Post-pruning first allows the tree to grow and then removes branches that contribute little predictive value.

Cost-complexity pruning can be used for this purpose.

Example comparison:

Model	Accuracy	Complexity
Pre-pruned Tree	86%	Low
Unpruned Tree	88%	High
Post-pruned Tree	89%	Moderate

The post-pruned Decision Tree would be preferred if it maintains or improves test performance while reducing unnecessary complexity.

For clinical deployment, a simpler pruned tree is preferable because it is easier to interpret, validate and explain to healthcare professionals.

Task 3: Statistical Learning for Risk Stratification

(a) Naïve Bayes / Logistic Regression

A statistical learning model can also be trained using the same dataset.

For example, Logistic Regression estimates the probability of diabetes using patient features.

The logistic function is:

$$P(Y=1) = 1 / (1 + e^{-z})$$

where:

$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

Example comparison

Model	Accuracy	AUC-ROC
Decision Tree	89%	0.91
Logistic Regression	87%	0.90

Decision Tree: Better at representing nonlinear relationships and interactions between features.

Logistic Regression: Easier to interpret and provides probability-based risk estimates.

A statistical model is preferable when:

- Interpretability is important.
- Relationships are relatively simple.
- Probability estimates are required.
- The dataset is not extremely complex.
- Clinical professionals need an easily explainable model.

(b) Feature Importance Analysis

Example top predictors:

Decision Tree

1. Glucose Level
2. BMI
3. Age

Logistic Regression

1. Glucose Level
2. BMI
3. Family History

Comparison

Both models identify Glucose Level and BMI as important predictors. The third predictor may differ because the two models calculate feature importance differently.

The Decision Tree uses reductions in impurity/information gain, while Logistic Regression uses the magnitude and direction of its coefficients.

Therefore, some agreement is expected, but complete agreement is not necessary.

Task 4: Reinforcement Learning for Treatment Recommendation

The healthcare treatment problem can be modeled as a Markov Decision Process (MDP). The assessment specifically requires states, actions, rewards, transitions and Q-Learning.

(a) MDP Model

States

Patient health stages can be represented as:

- **S1 – Low Risk**
- **S2 – Moderate Risk**
- **S3 – High Risk**
- **S4 – Critical/Uncontrolled**

A state can be determined from glucose, BMI, blood pressure and other patient measurements.

Actions

The available actions are:

- **A1 – Diet**
- **A2 – Exercise**
- **A3 – Medication**
- **A4 – Monitor**

Reward Function

A simple reward scheme can be:

Patient Outcome	Reward
Significant health improvement	+10
Moderate improvement	+5
No significant change	0
Health deterioration	-10

The reward encourages the agent to select actions that result in better patient health.

Transition Dynamics

The transition function:

$$P(S' | S, A)$$

represents the probability of moving from current state S to the next state S' after performing action A .

For example:

Moderate Risk

|

| Exercise

↓

Low Risk

or

High Risk

|

| Medication

↓

Moderate Risk

In an actual medical system, transition probabilities should be learned or validated using appropriate clinical data rather than invented.

(b) Q-Learning

Q-Learning learns the value of performing an action in a particular state.

The update equation is:

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

Where:

- α = learning rate
- γ = discount factor
- r = reward
- s = current state

- a = current action
- s' = next state

Assume:

$$\alpha = 0.5$$

$$\gamma = 0.9$$

Initially all Q-values are 0.

Iteration 1

Suppose:

$S2 \rightarrow \text{Exercise} \rightarrow S1$

Reward = +10

Therefore:

$Q(S2, \text{Exercise})$

$$= 0 + 0.5[10 + 0.9(0) - 0]$$

$$= 5.0$$

Another transition:

$S3 \rightarrow \text{Medication} \rightarrow S2$

Reward = +5

$Q(S3, \text{Medication})$

$$= 0 + 0.5[5 + 0.9(0) - 0]$$

$$= 2.5$$

Another:

$S1 \rightarrow \text{Monitor} \rightarrow S1$

Reward = +5

$Q(S1, \text{Monitor})$

$$= 0 + 0.5[5 + 0.9(0) - 0]$$

$$= 2.5$$

Iteration 2

Suppose:

$S2 \rightarrow \text{Exercise} \rightarrow S1$

Current:

$$Q(S2, \text{Exercise}) = 5$$

Maximum Q-value in $S1 = 2.5$

Therefore:

$$Q(S2, \text{Exercise})$$

$$= 5 + 0.5[10 + 0.9(2.5) - 5]$$

$$= 7.625$$

For:

$S3 \rightarrow \text{Medication} \rightarrow S2$

Current $Q = 2.5$

Maximum Q-value in $S2 = 7.625$

$$Q(S3, \text{Medication})$$

$$= 2.5 + 0.5[5 + 0.9(7.625) - 2.5]$$

$$= 6.1875$$

For:

$S1 \rightarrow \text{Monitor} \rightarrow S1$

Current $Q = 2.5$

Maximum Q-value in $S1 = 2.5$

$$Q(S1, \text{Monitor})$$

$$= 2.5 + 0.5[5 + 0.9(2.5) - 2.5]$$

$$= 3.875$$

Q-Table Example

State	Diet	Exercise	Medication	Monitor
S1 Low Risk	0	0	0	3.875
S2 Moderate	0	7.625	0	0
S3 High	0	0	6.188	0
S4 Critical	0	0	0	0

The agent should be trained for at least five patient episodes, as required by the assessment.

Convergence Criterion

Training can be considered converged when:

Maximum change in Q-values < 0.001 for several consecutive episodes

or when the learned policy remains unchanged for a predefined number of episodes.

(c) Final Learned Policy

The policy chooses the action having the highest Q-value for each state.

Example:

Patient State	Best Action
S1 – Low Risk	Monitor
S2 – Moderate Risk	Exercise
S3 – High Risk	Medication
S4 – Critical	Medication / Clinical escalation

Therefore:

Low Risk → Monitor

Moderate Risk → Exercise

High Risk → Medication

Critical → Clinical intervention

Important: In a real healthcare deployment, an RL agent should not autonomously prescribe or change medication. The learned policy would need clinical validation, safety constraints and human medical supervision.

Ethical Considerations

1. Patient Safety

Incorrect recommendations could harm patients. The system must have safety constraints and human oversight.

2. Bias

If the training data is biased or does not represent different patient groups adequately, the RL agent may produce unfair recommendations.

3. Explainability

Healthcare professionals should understand why an action was recommended.

4. Privacy

Patient records contain sensitive information and must be securely stored and processed.

5. Human-in-the-Loop

The AI should support doctors rather than completely replace clinical decision-making.

6. Validation

The system must be thoroughly tested and clinically validated before deployment.

Conclusion

The proposed system combines inductive learning, Decision Trees, statistical learning and reinforcement learning to support diabetes diagnosis and treatment recommendation. Decision Trees provide interpretable diagnostic predictions, while Logistic Regression/Naïve Bayes can provide statistical risk estimates. Q-Learning can model treatment decisions as sequential actions, but any real-world medical deployment must prioritize patient safety, fairness, explainability and clinician supervision.